

Handology assignment 1:

Matus Stanke

1.

a) draw 2 cards from 52 cards deck Without replacement. How many different pairs

$$\frac{52 \cdot 51}{2} = 2652 \text{ different pairs}$$

b) draw 2 cards from 52 Without replacement. (Do not care about order)

IF WE DO NOT CARE ABOUT ORDER, THEN PAIRS: $x, y = y, x$ so I WILL NOT COUNT THEM.

I SINCE WE draw 2 cards $\rightarrow$ JUST DIVIDE: $2652 \text{ PAIRS} / 2 = 1326$

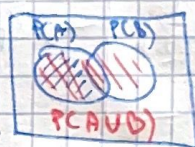
II OR use combinations not repeating formula:

$$\binom{n}{k}; \quad n \rightarrow \text{cards Total} \quad k \rightarrow \text{How many cards drawing} \rightarrow \binom{52}{2} = \frac{52!}{50!2!} = \frac{52 \cdot 51}{2} = 26 \cdot 51 = 1326$$

2 Events A, B

a) $P(A) \leq P(A \cup B) \wedge P(B) \leq P(A \cup B)$

TRUE



$P(A)$ will be always less or equal to $P(A \cup B)$
 $P(B)$ ————

b) $P(A) \geq P(A \cap B) \wedge P(B) \geq P(A \cap B)$

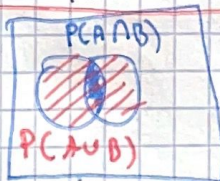
TRUE



$P(A)$ will be always more or equal than $P(A \cap B)$
 $P(B)$ ————

c) $P(A \cap B) \leq P(A \cup B)$

TRUE

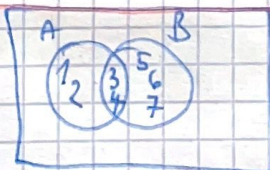


$P(A \cap B)$ will be always less or equal to $P(A \cup B)$

d) $P(A \cup B) = P(A) + P(B)$

$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6, 7\}$$



$7 \neq 9 \Rightarrow$ FALSE

$$P(A \cup B) = 1, 2, 3, 4, 5, 6, 7 \rightarrow 7$$

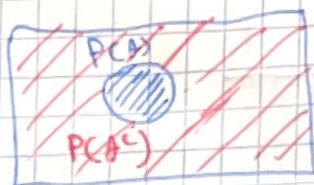
$$P(A) = 1, 2, 3, 4 \rightarrow 4$$

$$P(B) = 3, 4, 5, 6, 7 \rightarrow 5$$

$4 + 5 = 9$

(2) $P(A) + P(A^c) = 1$

TRUE



(3)

(a) toss coin 3 times. first 2 coins are tails?

$$\frac{T}{2} \cdot \frac{T}{2} \cdot \frac{H}{2} = \frac{1 \cdot 1 \cdot 1}{2 \cdot 2 \cdot 2} = \frac{1}{8} = \frac{1}{4}$$

(b) exactly 1 Head and other tails in n times

exactly one Head from $n \rightarrow \binom{n}{1} = \frac{n!}{(n-1)! \cdot 1!} = \frac{(n-1)! \cdot n}{(n-1)! \cdot 1} = \frac{n}{1} = n \text{ (1 Head)}$

$\rightarrow$ Tails $\left(\frac{1}{2}\right)^n \rightarrow \frac{T}{2} \cdot \frac{T}{2} \cdot \frac{T}{2} \cdots = \frac{1^n}{2^n} = \left(\frac{1}{2}\right)^n$

$\Rightarrow n \cdot \left(\frac{1}{2}\right)^n \rightarrow$ Probability of 1 Head and other tails

$\downarrow$ $\downarrow$
 1x Head tails

(c) throw dice 2x. get 6 both times

$$\frac{1 \cdot 6}{6 \cdot 6} = \frac{1}{36}$$